

# Sho Sakaue

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## EDUCATION

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### University of Utah

*Bachelor of Science in Computer Science and Applied Mathematics*

Salt Lake City, Utah

Aug. 2024 - Expected Dec. 2027

- **GPA:** 3.97/4.0
- **Relevant Coursework:**
  - \* **CS:** Machine Learning & Deep Learning, Probabilistic Deep Learning, Natural Language Processing, Computer Vision, Computational Geometry, Data Structures & Algorithms, Models of Computation
  - \* **Math:** Optimization, Probability Theory & Statistics, Linear Algebra, Numerical Analysis, Complex Analysis, Real Analysis, Calculus I-III
  - \* **Large Language Models:** - Matsuo Lab, University of Tokyo

June. 2026 - Present

## RESEARCH EXPERIENCE

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### Undergraduate Research Assistant - under Professor Suhas Prameela

*University of Utah*

Aug. 2025 - Present

*Salt Lake City, Utah*

- Generated and pre-processed training datasets for Machine Learning (ML) models for atomic simulation.
- Increased training convergence rate by day and model performance selecting optimal optimization method.
- increased model prediction accuracy by 15 % on average while maintaining inference speed.

## CONFERENCE & PRESENTATION

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### TMS 2027 Annual Meeting & Exhibition - Accepted for Oral Presentation

Mar. 2027

- Abstract accepted for oral presentation: *Particle shape and interface chemistry control defect relaxation in GRX-810 ODS alloys.*
- Co-authored research examining how particle morphology and oxide-matrix interface chemistry influence defect relaxation in GRX-810 oxide dispersion strengthened alloys.

### Undergraduate Research Symposium - University of Utah

July. 2026

- Delivered a presentation on research funded through the Undergraduate Research Opportunity Program (UROP).
- Presented research on developing *machine learning interatomic potentials for NiCoCr-based alloys.*

## WORK EXPERIENCE

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### Computer Science Teaching Assistant - CS 1400 (Intro to Computer Programming)

*University of Utah*

Aug. 2026 - Present

*Salt Lake City, Utah*

- Led weekly group lab session to help students practice coding fundamentals and clarify concepts beyond lectures.
- Graded coding assignments for 100+ students based on correctness, readability, and programming best practice.
- Assisted students in debugging programs and developing their understanding through office hours.

### Mathematics Tutor and Grader

*University of Utah*

Aug. 2025 - July. 2026

*Salt Lake City, Utah*

- Tutored students in Calculus I, II, III, Linear Algebra, and Statistics through clear explanations for each student.
- Worked with 10+ professors to grade 1,000+ student submissions, providing detailed feedback.
- Adapted strategies to each student's learning style to improve understanding and problem-solving independence.

PROJECTS

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| <b>BayesBench</b>   <i>numpy, scikit-learn, matplotlib, pandas</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | June. 2026 - Present  |
| <ul style="list-style-type: none"><li>Implemented the benchmark of Bayesian Optimization method for various limited-budget black-box optimization.</li><li>Added experiment scripts and plotting tools to compare optimization performance across methods.</li><li>Structured the project as a reusable Python package for future extension to more benchmarks and optimizers.</li></ul>                                                                                                                                  |                       |
| <b>LoRA and QLoRA for Fine-Tuning</b> (Paper reproduction)   <i>PyTorch, Hugging Face</i>                                                                                                                                                                                                                                                                                                                                                                                                                                 | Mar. 2026 – Apr. 2026 |
| <ul style="list-style-type: none"><li>Reproduced the LoRA and QLoRA methods from their original research papers, translating the proposed low-rank adaptation and quantized fine-tuning techniques into a working training pipeline.</li><li>Fine-tuned a llama-2-1B and benchmarked LoRA and QLoRA against the base model and full fine-tuning, comparing accuracy-performance tradeoffs.</li><li>Conducted hyperparameter experiments across LoRA ranks and quantization bit widths to analyze their effects.</li></ul> |                       |
| <b>Multiplayer Networked Game &amp; HTTP Server</b> - <i>Software Practice</i>   <i>C#, SQL</i>                                                                                                                                                                                                                                                                                                                                                                                                                           | Nov. 2025 - Dec. 2025 |
| <ul style="list-style-type: none"><li>Developed a distributed multiplayer game client in Blazor using the MVC architectural pattern.</li><li>Implemented a multi-threaded TCP network controller to communicate by JSON, using locks to ensure thread safety and prevent race conditions.</li><li>Integrated a MySQL database to store game history and player high scores.</li></ul>                                                                                                                                     |                       |

TECHNICAL SKILLS

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|-------------------------------------------------------------------------------------------------|
| <b>Programming Languages</b> : Python, C/C++, Java, Bash, SQL(MySQL), C#                        |
| <b>Markup &amp; Style Languages</b> : HTML, CSS, L <sup>A</sup> T <sub>E</sub> X, Markdown      |
| <b>Machine Learning</b> : PyTorch, NumPy, Hugging Face, scikit-learn, SciPy, Matplotlib, Pandas |
| <b>Developer Tools</b> : Git(GitHub, GitLab), Docker, Linux, HPC/SLURM computing                |

COMPETITION/HACKATHON

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| <b>MIT iQuHACK</b> (Quantum Computer)                                                                                                                                                                                                                                                                                                                                                                                | Feb. 2026 |
| <ul style="list-style-type: none"><li>Simulated 50–60 qubit quantum circuits using Qiskit to identify hidden peak bitstrings, placing 24th in BlueQubit Challenge.</li><li>Optimized quantum circuits simulation performance by gate reordering, reducing runtime by approximately 30%.</li><li>Developed statistical candidate-selection heuristics to infer peak from near-uniform output distributions.</li></ul> |           |
| <b>Crimson Hack</b> (University of Utah)                                                                                                                                                                                                                                                                                                                                                                             | Oct. 2025 |
| <ul style="list-style-type: none"><li>Built a rocket simulation game, where players navigate from one planet to another while avoiding obstacles.</li><li>Implemented planet-specific gravity and fuel-limited movement mechanics considering physical reality.</li><li>Designed interactive gameplay logic that allowed users to select different solar-system planets and control rocket</li></ul>                 |           |

AWARDS & CERTIFICATES

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|---------------------------------------------------|-------------------------|
| Frank Stenger Scholarship                         | Fall 2026               |
| Undergraduate Research Opportunity Program (UROP) | Summer 2026             |
| 24 th place in BlueQubit challenge in MIT iQuHack | Feb. 2026               |
| Dean’s List                                       | Fall 2024 - Summer 2026 |
| ISSS Scholarship                                  | Spring 2026             |
| Utah Global Scholarship                           | Fall 2024 - Fall 2027   |

LANGUAGE

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| <b>English</b> - Fluent  |
| <b>Japanese</b> - Native |